LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Chemical Reactions and Equations

Types of reactions

Type	What happens	Example
Combination	Two or more reactants form one product	$CaO + H_2O \rightarrow Ca(OH)_2$
Decomposition	One reactant breaks into simpler products (by heat, electricity or light)	$CaCO_3 \rightarrow CaO + CO_2$ (on heating)
Displacement	A more reactive element pushes a less reactive one out of its compound	$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
Double displacement	Two compounds exchange ions, often forming a precipitate	$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$
Redox	One substance is oxidised while another is reduced	$CuO + H_2 \rightarrow Cu + H_2O$

Must-know points

- **Oxidation** = gain of oxygen or loss of hydrogen. **Reduction** = loss of oxygen or gain of hydrogen.
- **Exothermic** reactions release heat (respiration, burning of fuels). **Endothermic** reactions absorb heat (most decomposition reactions).
- Signs of a chemical reaction: change in state, change in colour, gas given off, change in temperature.
- Balance equations with **coefficients only**. Never change the subscripts in a formula.

Corrosion: a metal is slowly eaten away by moisture, air or acids. Rust = $Fe_2O_3 \cdot xH_2O$. Prevent with paint, oil/grease or galvanising.
Rancidity: fats and oils get oxidised and their smell and taste change. Prevent with antioxidants, nitrogen flushing, airtight containers or the fridge.

Common mistakes

- Changing a formula ($H_2O \rightarrow H_2O_2$) to "balance" an equation.
- Forgetting physical states (s), (l), (g), (aq) when the question asks for them.
- Writing "blue to green" for $Fe + CuSO_4$. The solution turns from blue to **light green**.

Quick check

1. Balance: $Al + O_2 \rightarrow Al_2O_3$

2. In $ZnO + C \rightarrow Zn + CO$, which substance is oxidised?

3. Why is respiration called an exothermic reaction?